Data Visualization & Analytics Lab

Experiment No. : 4

Design custom visualizations using filters and visual enhancement techniques

Experiment No.4

1. **Aim:** Design custom visualizations using filters and visual enhancement techniques.

2. **Objectives:**

   - Understand why visualization is an important part of data analysis.
   - Understand the components involved in visualization design.

3. **Course Outcomes:** Students will be able to apply various filtering and customization options to enhance data visualization and analysis.

4. **Hardware / Software Required: PowerBI/Tableau**

5. **Theory:**

Interactive Visualization -

Interactive visualization allows users to interact with visual data elements. Instead of viewing static charts, users can modify the visualization by selecting filters or clicking specific elements. Interactive dashboards allow users to explore data from different perspectives.

For example, a user can filter sales data by region, category, or year to analyze performance more effectively. This flexibility makes interactive visualization an important component of modern data analysis tools.

Filters in Visualization Tools

Filters are used to restrict or display specific subsets of data. They allow users to focus only on the information relevant to their analysis.

Different types of filters include:

1. Basic filters that allow users to select specific categories or values.
2. Range filters used for numerical data such as sales or profit.
3. Date filters used to analyze data for specific time periods.
4. Slicers which allow users to dynamically control data displayed in dashboards.

Visual Enhancement Techniques

Visual enhancement techniques improve the readability and effectiveness of dashboards. These techniques make the visualization more appealing and easier to understand.

Common visual enhancement methods include:

1. Conditional formatting to highlight important values using colors.
2. Tooltips that display additional information when the user hovers over a chart element.
3. Color themes to maintain visual consistency across the dashboard.
4. Labels and annotations to explain important insights.
5. Interactive highlighting which updates other charts when one visual is selected.

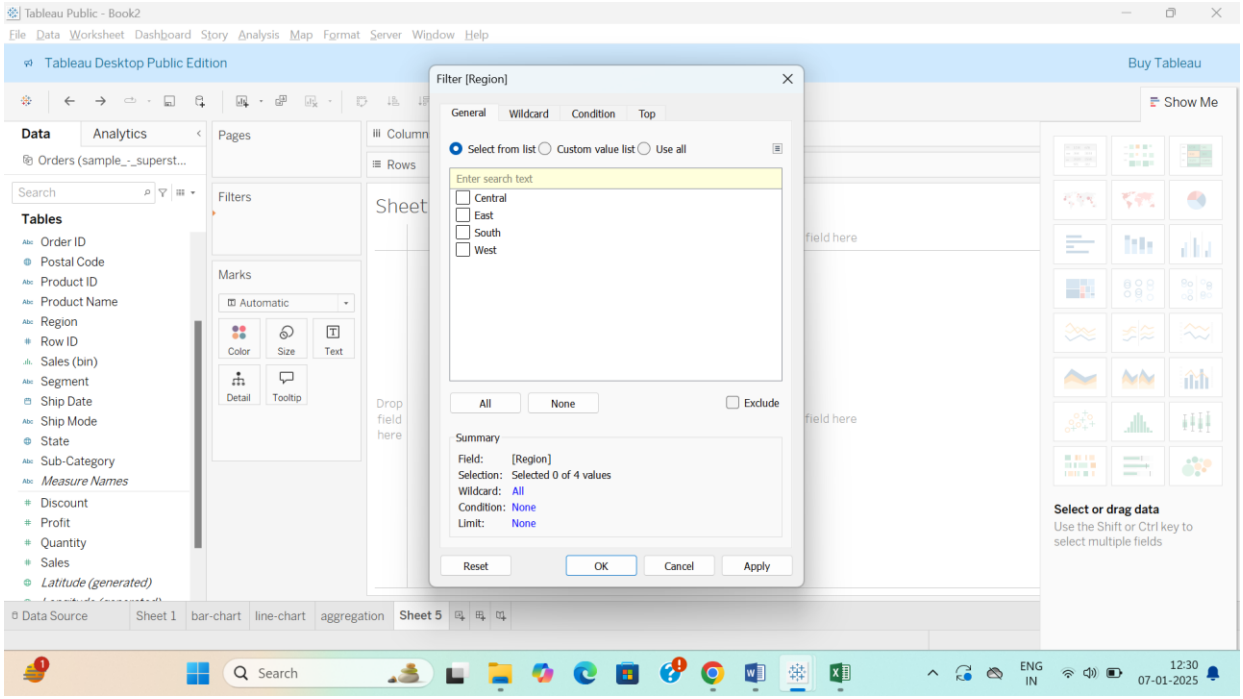
Implementation in Tableau:

Filters in Tableau:

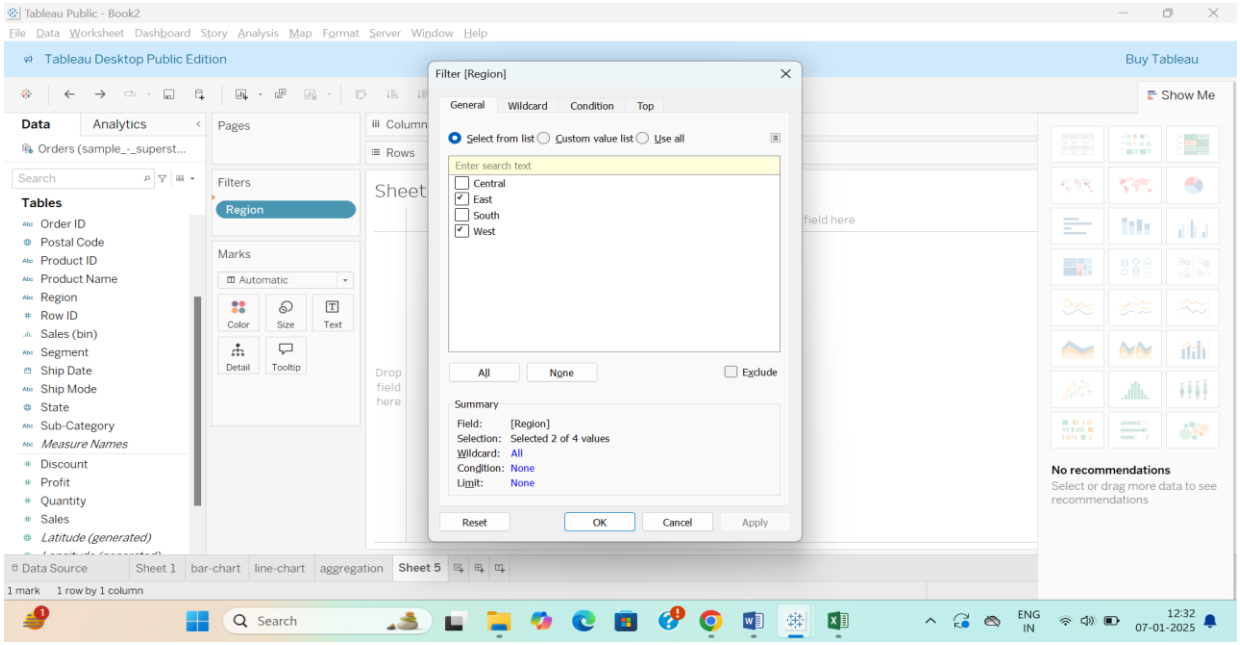
- **Dimension Filters:** Filters based on categorical data (like product names, regions, or customer IDs)
- **Measure Filters:** Filters based on quantitative data (e.g., sales, profit, temperature).
- **Date Filters**: Filtering by year, quarter, month, week, or specific date ranges
- **Top N / Conditional Filters:** Filters to display only the top or bottom N items based on a specific measure.
- **Context Filters:** A context filter acts as an independent filter that sets the context for other filters.
- **Wildcard Filters:** Filters that allow you to apply wildcard matching on dimensions, providing more flexible filtering options. E.g. Filtering for product names that contain a specific keyword.
- **Data Source Filters:** Filters applied directly to the data source level, restricting the data loaded into Tableau. E.g. Filtering out rows from a database that are irrelevant to the analysis.

Setting Up Filters in Tableau:

- Drag a field (e.g., Region or Sales) to the Filters shelf.
- A dialog box will appear to configure the filter.



Example: Filter by Region to display only "East" and "West" regions



Students can apply various colors, labels, and tooltips to enhance the visual appearance of charts.

In similar way, Students can try the filtering and customization in PowerBI.

5. **Conclusion:** By using filters, slicers, and visual enhancement techniques, users can explore data more effectively and gain valuable insights.

6 Viva Questions:

- What are different types of filters used to enhance data visualization?
- What is interactive visualization?
- What are different visual enhancement techniques?

References:

1. Filter Data from Your Views: https://help.tableau.com/current/pro/desktop/en-us/filtering.htm
2. Types of filters in Power BI reports: https://learn.microsoft.com/en-us/power-bi/create-reports/power-bi-report-filter-types